GameAHN Project Report

Haydn Gamblin, Alex Henderson, and Nathanael Tagert

Department of Computer Science, Mississippi College

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Dr. Glenn Wiggins

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Introduction

The purpose of this project was to provide a small gaming website for users to play games and keep record of their wins and losses for a good, fun experience. This type of website is great for “blowing off steam” and relaxing, especially after a hard day’s work!

This project, created by the efforts of Nathanael Tagert, Haydn Gamblin, and Alex Henderson is a simple website that includes a few games, linked to a database that adds a user’s combined win and loss totals to a profile. The available games include Solitaire, Tic-Tac-Toe, Checkers, and Connect-4, with the latter three having both local multiplayer and versus computer modes.

We organized ourselves using a main, shared Google Document. This document contained our Scrum meeting record, links to the open source code we implemented, and the main colors we were to use in our user interface and design. We also included notes following individual links for potential changes that should be considered upon or after implementation that we removed post-fixes. VER was the abbreviation for version, and was used to distinguish between the code for computer versus games and local multiplayer games. Users may also be referred to as players, considering the project is focused on providing an enjoyable experience through games.

In the case that this product continues development, much expansion is possible in the form of an increased game library, more extensive statistical records, and quality-of-life features based on user feedback.

Project Management Plan

We implemented a simple DevOps lifecycle process when developing our project. We began with a plan using open source versions of the games to speed up development time, a

CSS-based website to provide users with a streamlined experience, and a SQL database to keep track of profile information. We used GitHub to monitor and regulate changes to the project, as well as to compile the different functions, classes, and files that were necessary. This resulted in two GitHub repositories. The first repository provided the base of the project, and the second cemented the final steps of the process, implementation, and final testing.

Each of us used Visual Studio Code to some degree, mainly focussing on the HTML, CSS, and JavaScript programming languages to ensure quick website development and implementation. Nathanael Tagert refreshed his knowledge of HTML, used JavaScript and CSS for the first time, and experienced open source code management for the first time. Hayden experimented with SQL implementation into a web application. Alex fully cemented her HTML, JavaScript, and CSS abilities with her main focus on the GUI elements and system. All project developers had never used GitHub before, and learned a substantial amount about the DevOps software scheme.

Requirement Specifications

The stakeholders for the project were the players. This game website project was created for their enjoyment, therefore, it was imperative they be pleased with the results.

The web page displays the games and user interface to the players so they may interact with the different games and access their profile statistics. Players enter their desired information to create a profile that is stored in the database for future management purposes through tallies of wins and losses.

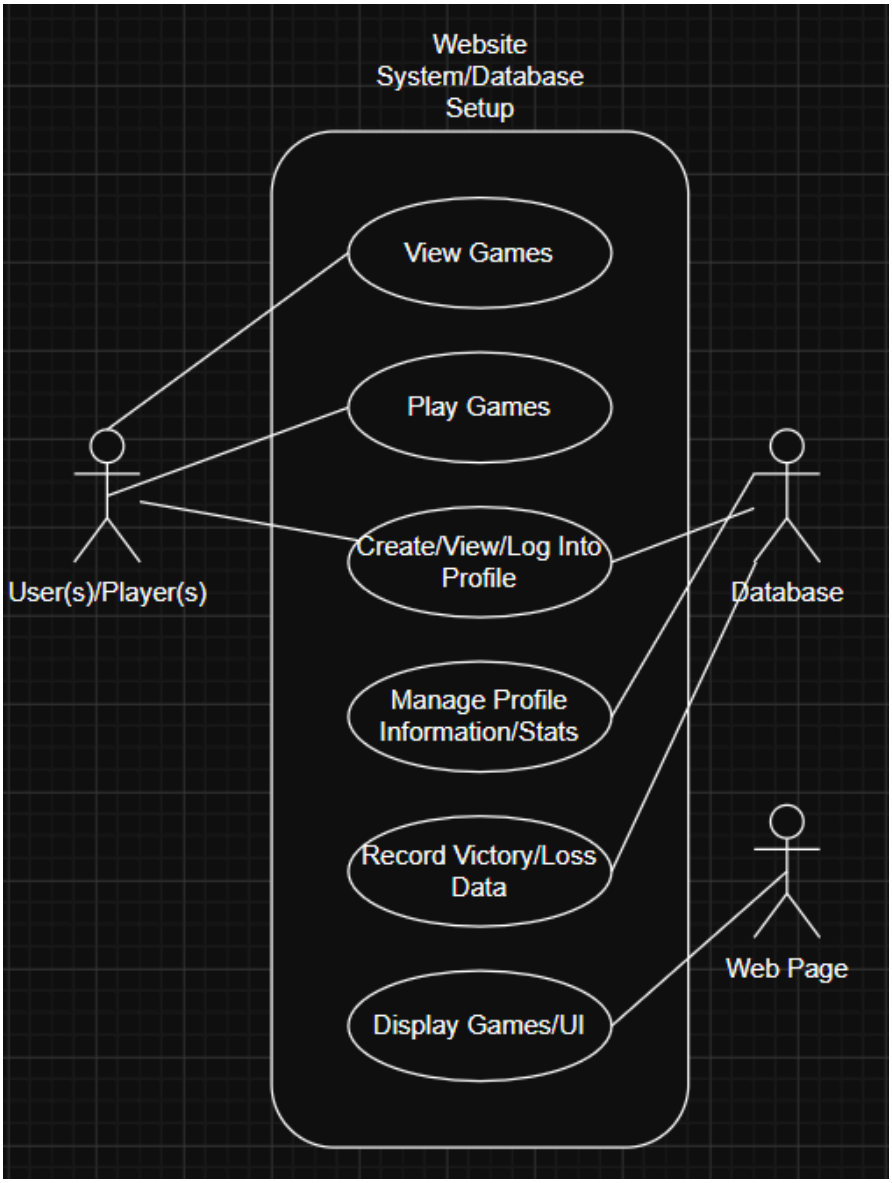

Figure #1: Graphic Use Case Diagram

Architecture

Our project implemented a vast array of technology in its creation. We used the Visual Studio Code software to individually work on code for the site. We also used the SQL, JavaScript, CSS, and HTML programming languages to configure the frontend and backend of our project. Finally, we used the web-based development tools offered by GitHub to focus our individual efforts into one, and also to find open source code for faster development. Without tools like these, website development would be a nearly impossible task.

Design

Our GUI begins at our main page that has our site name, GameAHN, which is a play on words that includes the initials of each of our names, while also making the statement "Game On!" The main page also contains log-in, sign-up, and profile buttons that direct users to different pages that allow them to create and view their profile information. Under the top banner lie our games, which are 4 highlightable buttons that link you to each of the games individually. The bottom banner contains our copyright date, names, and other information to identify our site's identity.

The overall design of our site is based on a black, green, and blue color scheme. This scheme is reflected in each of our games, with game pieces consisting of some combination of these colors. Each game page includes a back button to return to the main page and a restart button to reset the game state. All games other than solitaire also include an AI button and a local button to determine what mode the user would like to play, whether they would like to play against a computer AI or another human locally.

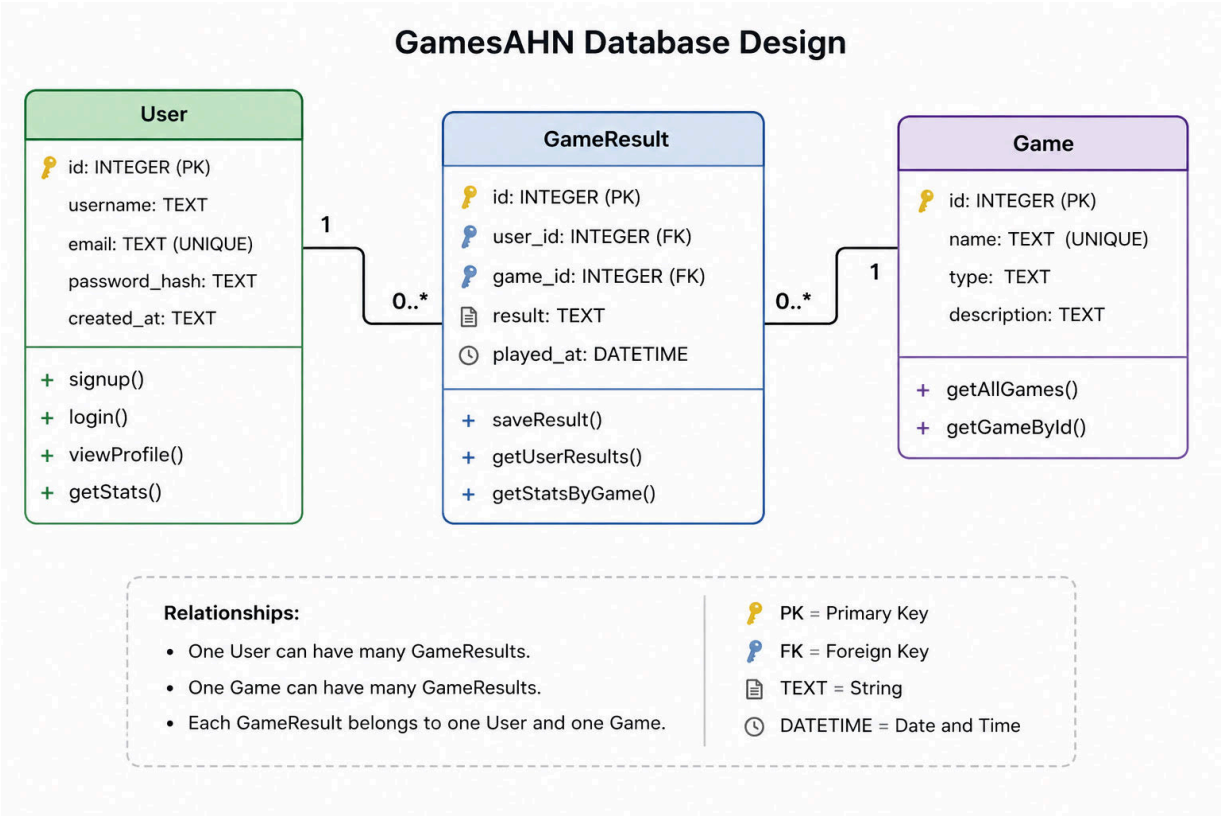

Figure #2: Database Class Diagram

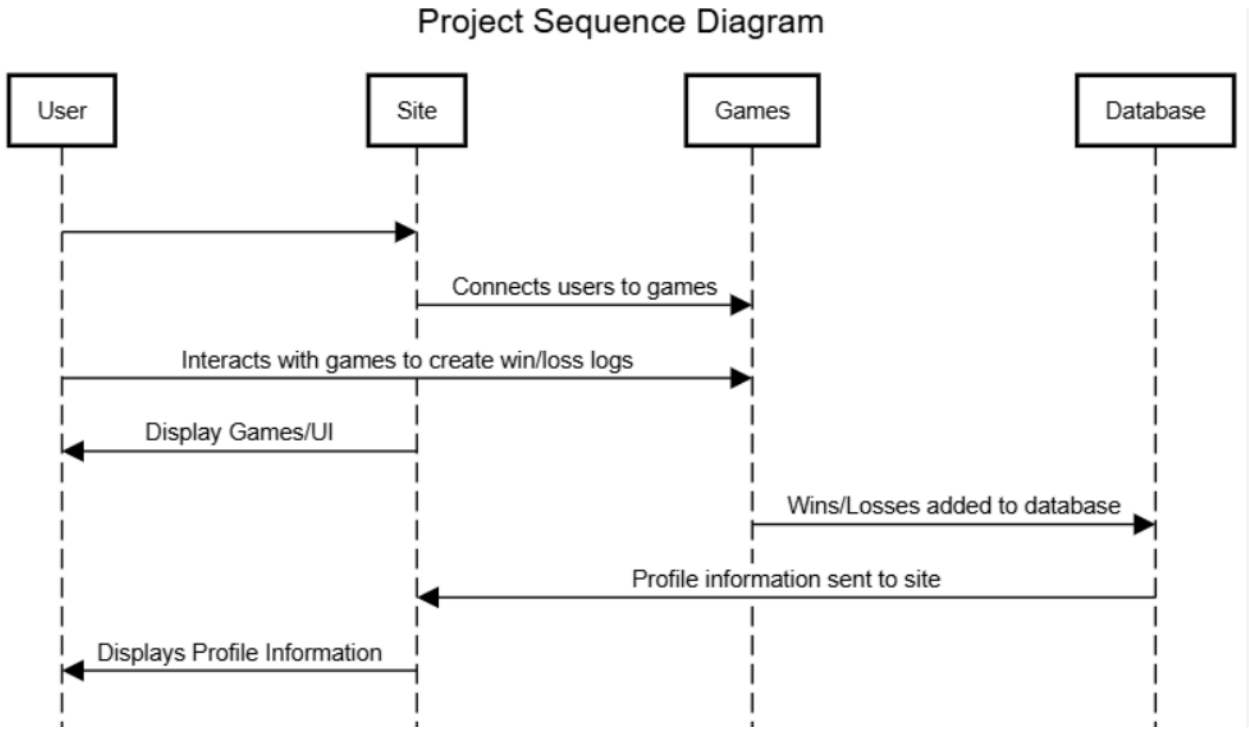

Figure #3: Project Sequence Diagram

Test Plan

Our test plan was a manual, user-based continuous testing scheme that involved our team personally testing each feature both before and after implementation to ensure proper functionality. This allowed us to test the games as if they were being played by an end user, while also getting some enjoyment from the results. After implementation testing and ensuring that all the pieces of the project were fitting together nicely, we requested others not initially involved in our project to also play the games and see if any changes were necessary. Because of the nature of a gaming website, being that users are playing games for fun, user testing was crucially necessary before the project could truly be called complete.

Open Source Code Used

AIMAN2244. (n.d.). *Solitaire game* [Source code]. GitHub. https://github.com/AIMAN2244/Solitare-Game.

Asudepeker. (n.d.). *Checkers game AI* [Source code]. GitHub. https://github.com/asudepeker/checkersGameAI.

Mostafa-Samir. (n.d.). Tic-tac-toe AI [Source code]. GitHub. https://github.com/Mostafa-Samir/Tic-Tac-Toe-AI.

BanhmiDev. (n.d.). *Connect four JS* [Source code]. GitHub. https://github.com/BanhmiDev/connect-four-js.

Coreyladovsky. (n.d.). *Connect4* [Source code]. GitHub. https://github.com/coreyladovsky/connect4.

Juancarloschou. (n.d.). *Game checkers* [Source code]. GitHub. https://github.com/juancarloschou/GameCheckers.

XrsQugi. (n.d.). Tic-tac-toe [Source code]. GitHub. https://github.com/xrsQugi/tic-tac-toe.